Hamed Hamzeh

Tehran, Iran
(+98) 9362998950
hamed.hamze2212@gmail.com
hamed-hamzeh-383660247
www.hamedhamzeh.github.io

Research Interests

- System Design and Optimization
- Numerical Modeling and Simulation
- Data Science
- Machine Learning
- Robotics

Education

2018–2023 **B.Sc. in Mechanical Engineering**,
University of Tehran, Tehran, Iran, GPA: 16.13/20, last two years GPA: 3.48/4

B.Sc.'s Thesis, *Design and development of a whole body continuous passive motion (CPM) device for neurorehabilitation. Under the supervision of Dr. Daneshmehr (18.50/20)*

Selected Courses

- Artificial Intelligence: 19.5/20
- Mechatronics: 16.75/20
- Dynamics: 19/20
- Fundamental of Electronics: 20/20
- Control: 16.3/20

Publications

Accepted **Design of a Remote Controlled Puppet Robot Imitating a manual Driven Puppet Using Deep Learning Pose Detection**, *ICROM 2024*,
H Hamzeh, K Shahroozi, ... , and H Moradi

Under Review **Design of A Bio-inspired, Compliant, and Modular Robot with Magnetic Adhesion System for Iron Surface Inspection**, *IEEE*,
P Parhami, M Salehpour, **H Hamzeh**, R Nasiri and H Moradi

Experience

Sep 2023 - present **Research assistant**, *ARIS Lab*, School of Electrical and Computer Engineering, University of Tehran
Supervisor: Prof. Manouchehr Moradisabzevar

Hand Puppeteer Robot: Visit webpage

- Engineered a wireless-controlled, 3D-printed robot, integrating design and coding for interactive movement.
- Programmed NodeMCU and Arduino to process real-time data from a gyroscopic sensor for robotic control.
- Investigated gesture-controlled robotics and selected **YOLO v8** for puppet detection and pose recognition.
- Developed and annotated a dataset of 2,000 images, including keypoints and bounding boxes.
- Achieved **91.4%** accuracy with the YOLO v8 pose detection model.
- Utilized **Kalman filtering** during inference to enhance the stability of pose estimation.
- Synced the pose detection model with the robot for remote-controlled movement.

Silkworm Robot: Visit webpage

- developed a modular, semi-soft robotic system for movement on metallic surfaces, aimed at inspection and maintenance applications.
- Designed and 3D-printed components using **PLA and TPU**, with additional parts made from Plexi.
- Developed a web-based interface for remote controling, utilizing ESP32 microcontroller.
- Achieved a **6.7%** lower cost of transportaion through experiments with variable module speeds.
- Co-authoring an upcoming publication detailing the design methodology, experimental findings, and potential applications of the Silkworm Robot in industrial inspection tasks.

Jul 2022 - Jun 2023 **Mechanical Engineer**
GANJE, a startup in the field of smart logistics

- Worked for 6 months in the Research and Development team to design and manufacture smart lockers.
- Developed new solutions to improve lockers' functionality and user experience through research and testing.
- Created CAD designs for the lockers, turning concepts into detailed models and drawings.
- Utilized a top-down design approach to efficiently manage complex assemblies.
- Spent another 6 months as a front-end developer intern, working on the website and webApp.
- Gained experience with HTML, CSS, JavaScript, React, and GitHub-based collaboration.

Jun - Aug 2021 **Internship**
Avita Company
- Collaborated with a team to design and manufacture wheelchairs using composites.
- Assisted in the manufacturing process, creating fixtures to enhance the assembling process.

Selected University Projects

Pediatric Bone Age Prediction Using Xception Model,,
- Predicted pediatric bone age using hand X-ray images with an Xception model.
- Applied data augmentation and tracked model preformance via Wandb, achieved a training error of 4.9 months and validation error of 9.1 months.

Exploration of Recurrent Networks: RNNs and LSTMs
- Studied and applied RNNs and GRUs to weather forecasting, with GRUs providing superior performance compared to RNNs.
- Developed and trained LSTM networks to overcome RNN limitations in capturing long-term dependencies; applied the model to analyze and forecast trends in S&P 500 stock market data.

Mechanical Component Classification with Tree Base models
- Data preprocessing, feature selection, and visualization to enhance model input.
- Trained different tree-based models, with hyperparameter optimization using Optuna.

Skills

Programming Python, Matlab, C++, Latex, HTML, CSS, JavaScript
Frameworks Scikit-learn, OpenCV, TensorFlow, Keras, React
Databases MySQL

CAD/CAE SolidWorks, Siemens NX, 3D print software, COMSOL
Platforms NodeMCU, Arduino, Git, Linux

Soft Skills Critical thinking, R&D, teamwork, Problem solving

Languages

Fluent English: IELTS: 7/9 R:7.5, L:8, S:7, W:6
Native Farsi

Certificate

May 2023 **Machine learning Specialization**, *link*
Institute: coursera
- Supervised Machine Learning: Regression and Classification
- Advanced Learning Algorithms
- Unsupervised Learning, Recommenders, Reinforcement Learning

References

Pr.Manouchehr Moradisabzevar, *Professor at College of Electrical and Computer Engineering*, University of Tehran, Email: moradih@ut.ac.ir
Relationship: Research Supervisor

Dr.Alireza Daneshmehr, *Assosiate Professor at College of Mechanical Engineering*, University of Tehran, Email: daneshmehr@ut.ac.ir
Relationship: Bachelor Thesis Supervisor

Dr.Rezvan Nasiri, *Assosiate Professor at College of Electrical and Computer Engineering*, University of Tehran, Email: rezvan.nasiri@ut.ac.ir
Relationship: Research Supervisor